MIL-C-10085E
7 April 1977
SUPERSEDING
MIL-C-10085D
15 April 1971

MILITARY SPECIFICATION

CUTTER, PHOTOGRAMMETRIC,

SLOTTED-TEMPLET: HAND-OPERATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a hand-operated punch with chest. The punch accurately die cuts radial slots about a common center hole.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple Wall.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

MIL-C-[illegible]

MILITARY

MIL-P-[illegible]	- Preservation-Packaging, Methods of.
MIL-P-5[illegible]	- Plates, Identification, Instruction and Marking, Blank.
MIL-S-[illegible]197	- Stud, Photogrammetric, Slotted Templet.
MIL-S-14249	- Straightedges, Radial-Line, Slotted Templet.
MIL-P-[illegible]250	- Pin, Straight, Headed and Pin, Straight, Headless, Slotted Templet Stud.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking For Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

DRAWINGS

ME

TA13217E7552	- Cutter, Slotted Templet, Hand-Operated.
TA13217E7715	- Chest, Cutter, Slotted Templet, Hand-Operated, With Accessories.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The cutter shall be in accordance with Top Assembly TA13217E7552, and as specified herein. The cutter shall cut or punch clean-cut center holes and slots without feathering their edges.

3.1.1 Chest. The chest shall be in accordance with Top Assembly TA13217E7715, and as specified herein.

3.1.2 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the cutter. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

3.2 First article (preproduction model). The contractor shall furnish one or more cutters as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce cutters that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.4 Alinement. The extended center lines of all slots punched or cut by the cutter shall not vary from the intersection at the center of the punched hole by more than 0.010 inch. The sides of all slots shall be parallel, and the width of all slots shall equal the diameter of the center hole.

3.5 Identification marking. The cutter shall be identified in accordance with MIL-STD-130. The marking shall be applied to the cutter on plates conforming to MIL-P-514, Type I, Style 3, Composition C, of Type I, Grade A, Class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.6 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.7 Accessories. Unless otherwise specified (see 6.2), the accessories listed in Table I shall be supplied with each cutter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

TABLE I. Accessories.

Item	Applicable document	Quantity
Straightedge, 12-inch	MIL-S-14249	1 each
Straightedge, 30-inch	MIL-S-14249	1 each
Studs, brass	MIL-S-14197	7 gross
Pins, headless, slotted templet	MIL-P-14250, Type II	1 pound
Pliers, regular, 8-inch		1 pair
Screwdriver, cross-tip, 4-inch blade (No. 1)		1 each
Screw driver, flat-tip, 6-inch blade		1 each
Hammer, ball peen, 2-pound		1 each
Wrench, box and open end, 3/8-inch		1 each
Wrench, allen, 1/8-inch		1 each
Wrench, allen, 5/32-inch		1 each
Wrench, allen, 3/16-inch		1 each

3.8 Workmanship. All parts, components, and assemblies of the cutter which include castings, forgings, fabricated parts, stampings, and machined surfaces shall be clean and free from sand, dirt, fins, pits, sprues, scale, and other harmful extraneous material. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges and corners are required. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.8.1 Machine work. Tolerances and gages for metal fits shall conform to the limits specified herein and on applicable drawings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

4.1.1 Components and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction cutter shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. The preproduction cutter shall be tested as specified in 4.5.2. Failure of this test shall be cause for rejection or for performing the inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for major defects as specified in 4.5.1. AQL shall be 2.5 percent defective.

4.4.3 Tests. Each cutter shall be tested as specified in 4.5.2. Failure of this test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The cutter shall be examined as specified herein for the following major defects:

101. Nonconformance to drawings.
102. Parts, components, or accessories damaged, missing, or not functioning correctly.
103. Workmanship not as specified.

4.5.2 Alinement test. Each cutter shall be tested for the purpose of determining the accuracy with which the slot cutter die and the guide pin are positioned on a common centerline. Two sheets of Bristol board (3-ply or 5-ply) each approximately 3- by 21-inches, shall be used to construct templets. Place the Bristol boards individually in the cutter and punch a center hole approximately 1 inch from one end of the Bristol board and midway between the sides. Each Bristol board shall be marked to indicate the "punch" side. Place the center hole on the guide pin and punch one slot approximately midway between the board sides 18 plus or minus 1/32 inches from the center hole. Drive two headless pins, conforming to MIL-P-14250, Type II, into a tabletop on 1-11/16 plus or minus 1/32 inch centers. Place two studs, conforming to MIL-S-14197, on the pins. Place the slot of one templet, marked punch side up, over the two studs. Place the slot of the second templet, marked punch side down, over the same studs. The relative position of the holes in the two templets should be observed and their edge overlap measured with a 40-power shop microscope. A variance greater than 0.020 inch, measured between the edges of the two holes, shall constitute failure of this test. To confirm the "hole deviation measurement," additional templet sets may be punched, measured, and tabulated to produce an "average hole deviation measurement".

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

104.	Materials, methods, and containers not as specified for Level A or Level B. Each incorrect material, method, or container shall constitute one defect.
105.	Cutter and accessories not preserved as specified for Level A.
106.	Cushioning not provided to prevent movement in the chest for Level A.
107.	Strapping not as specified for Level A.
108.	Marking illegible, incorrect, incomplete, or missing for Level A, Level B, or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. The cutter, accessories, repair parts, and tools shall be preserved in accordance with MIL-P-116, Method I. Preservatives and their application shall be in accordance with the preservative application criteria of MIL-P-116. Technical publications shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3. The preserved items shall be placed in the chest provided as space permits. Items not fitting in the chest shall be consolidated in a close-fitting box conforming to PPP-B-636, W5c, style optional. Cushioning shall be provided to prevent movement or damage to the components contained within the chest or box. The chest lid shall be closed and secured. The box closure shall be as specified for Method V in the appendix to the box specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

5.1.2 Commercial. The cutter, accessories, repair parts, tools, and publications shall be preserved to afford protection against deterioration and damage during shipment from the contractor to the initial destination. The items shall be placed in the chest provided, as space permits. Items not fitting in the chest shall be consolidated in a close-fitting fiberboard box.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Each complete cutter preserved as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-621, Class 2, style optional, or in a box conforming to PPP-B-601, Overseas Type, Grade B, style optional. Each box shall be closed and strapped in accordance with the appendix to the applicable box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. Each complete cutter preserved as specified in 5.1 shall be packed in a box conforming to PPP-B-640, Class 2, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.3 Commercial. Each complete cutter preserved as specified in 5.1 shall be prepared for shipment in a manner to assure carrier acceptance and safe delivery to the initial destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall reflect as a minimum, the item name, national stock number, quantity, contract or purchase order number, and appropriate address markings on the shipping container. The marking shall be applied with a waterproof material, shall be legible, and shall provide a definite contrast.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The slotted-templet cutter is used in preparing, by punching or cutting, templets of cardboard or other suitable material from vertical serial photographs or rectified oblique trimetrogon aerial photographs in connection with the establishment of photogrammetric control by the radial line method. The chest is used for storage and transport of the cutter and accessories.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model and number of cutters required (see 3.2).
(c) Accessories required when other than as specified (see 3.7).
(d) Degree of preservation and packing required (see 5.1 and 5.2).
(e) When other than Finish B strapping is required (see 5.2.1).

6.3 Preproduction model. Any changes or deviations of production cutters from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish cutters conforming to this specification.

6.4 Data requirements. The contracting officer should include requirement for such data as technical publications, instructional material, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each cutter.

6.5 Provisioning. The contracting officer should include requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of cutters.

Custodians:
Army - ME
Air Force - 99

Review activity:
DLA - GS

Preparing activity:
Army - ME

Project 6675-0640

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-C-10085E Cutter, Photogrammetric, Slotted-Templet: Hand-Operated

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Mobility Equipment Research
and Development Command
ATTN: DRDME-DS
Ft. Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:32Z
Check the source to verify that this is the current version before use.